File 70: Post-Restoration As-Built Monitoring Report

Construction Verification & Spawning Habitat Success for Anderson Creek Reach

> ***Post-Restoration (As-Built) Project Standard**: This document represents the official post-work As-Built construction verification and monitoring report for the **Anderson Creek Showcase Restoration Project (Roya's Cabin Reach)**. It documents the successful geomorphic conversion of the 1,500 linear foot reach from its degraded pasture state into a stable, meandering, coldwater trout spawning habitat. It verifies that all NCD rock/log structures have been installed to design standards, and presents physical, thermal, and biological data proving the recovery of wild **Brown Trout (`Salmo trutta`)** spawning.*

I. Construction Completion & As-Built Verification

- **Project Name**: Anderson Creek Showcase Restoration (Roya's Cabin Reach)
- **Permit Number**: USACE Nationwide Permit 27 (SAS-2025-00918)
- **State Variance**: Georgia DNR EPD Buffer Variance Approved (BV-124-25-02)
- **Construction Period**: 12 Business Days (Completed October 2025)
- **As-Built Survey Date**: October 28, 2025

Construction Narrative

Blue Ridge Stream Restoration field crews successfully mobilized and executed instream grading and structure placement over a 12-day window in October 2025 during low seasonal baseflows. Using high-efficiency sediment pump-around bypass piping, instream grading was completed with zero sediment discharges downstream. The collapsing raw pasture banks were regraded to a stable 3:1 slope, wrapped in coir matting, and planted immediately. All geomorphic rock cross-vanes, log J-hooks, and bioengineered toe-wood structures were installed to precise geomorphic design coordinates.

```
                    RESTORED SPAWNING STREAM PROFILE (POST-WORK)

 [ stabilized bank ]                                     [ stabilized bank ]
  double-net coir                                         double-net coir
  & live stakes                                           & live stakes
      ■                                                        ■
      ▼        ■■■■■■■■■■■■■■■■■ Narrowed: 12.5 ft ■■■■■■■■■■■■■■■■       ▼
 _______                                                  _______
        \__                                            __/
           \__                                      __/
              \___  [ toe-wood layer ]           __/
                  \__________________________________/   ■■■ Thalweg pool depth: 4.5 ft
                      Sorted Clean gravel spawning runs
```

II. Geomorphic Transformation Data (Before vs. After)

Post-construction RTK GNSS monument cross-section surveys verify that the degraded, unstable pasture ditch (Rosgen G4) was successfully converted into a stable, meandering step-pool gravel-dominated stream (Rosgen B4c):

Geomorphic Comparison Ledger

Fluvial Geomorphic Parameter	Pre-Restoration (Degraded)	Post-Restoration (As-Built)	Design Target Standard	Net Improvement Class
Rosgen Classification	**G4** (Entrenched Gully)	**B4c** (Gravel Riffle-Pool)	**B4c**	Stable channel type
Bankfull Width (Wbkf)	24.2 feet	**12.5 feet**	12.0 feet	-48% channel narrowing
Mean Depth (dbkf)	1.0 feet	**2.2 feet**	2.4 feet	+120% pool depth increase
Max Thalweg Depth (dmax)	1.8 feet	**4.5 feet**	4.5 feet	Restored deep pool refuges
Width-to-Depth Ratio (W/D)	**24.2** (Overwidened)	**5.8**	5.0	Optimal bed shear velocity
Entrenchment Ratio (ER)	**1.15** (Entrenched)	**2.45** (Floodplain Access)	>= 2.2	Storm flows access floodplain
Sinuosity (K)	**1.02** (Straight Ditch)	**1.28** (Meandering Reach)	1.30	+25% meander path length
Median Substrate (D■■)	**2.4 mm** (Silt Clogged)	**22.0 mm** (Clean Gravel)	22.0 mm	Fine silts scoured clean

III. Physical & Biochemical Spawning Recovery Metrics

Continuous telemetry monitoring and field sampling completed in November 2025 during the primary brown trout spawning window prove the ecological success of our geomorphic restoration:

```
             RESTORED THERMAL & BIOCHEMICAL PROFILE (JULY)

[ MATURE CANOPY SHADE ] ■■■> [ 64.2°F COOL SUMMER COLD ] ■■■> [ ZERO THERMAL STRESS ]
                                                                                  ■
                                                                                  ▼
[ WATER COLUMN DO (8.6 mg/L) ] <■■ [ SATURATION ELEVATION ( Henry's ) ] <■
```

1. Summer Thermal Cooling Performance

Continuous streamside temperature sensors recorded a maximum summer water temperature of **64.2°F (17.9°C)** inside the restored reach on July 14, 2026.

- *Ecological Benefit*: Compared to the unshaded baseline reach (71.8°F), the restored canopy and deep stratified plunge pools cooled the water by **7.6°F**, maintaining stream temperatures well below the lethal limit for brown trout and providing a crucial coldwater refuge.

2. Intragravel Dissolved Oxygen (DO) Saturation

Intragravel DO probes placed 15cm deep into the restored gravel spawning riffles recorded an average dissolved oxygen level of **8.6 mg/L** (compared to the baseline hypoxic 5.2 mg/L).

- *Ecological Benefit*: Exceeds the state and biological brown trout egg survival threshold of **6.0 mg/L** by over **43%**, ensuring a high hatching success rate for developing trout embryos.

3. Stokes' Law Silt Reduction & Gravel Sorting

A post-work Wolman pebble count and dry-sieve analysis of the spawning riffles verified a clean substrate:

- **Silt/Sand Fraction (<2mm)**: Reduced from 58% down to **less than 5%**.
- **The Geomorphic Process**: Constricting the channel bankfull width from 24.2 ft to 12.5 ft concentrated hydraulic shear stress. High-velocity storm flows successfully scoured and transported fine silts downstream, leaving a beautifully sorted bed of clean, gravelly substrate (D■■ = 22mm) ideal for nest excavation.

IV. Biological Recovery: Benthic Macroinvertebrates

A biological assessment was executed in October 2026, one year following construction, using the EPA Rapid Bioassessment Protocol (RBP) III:

- **Total Taxa Richness**: Increased from 12 to 28 distinct aquatic insect families.
- **EPT Index (Ephemeroptera, Plecoptera, Trichoptera)**: Elevated from **2 to 14**!
- *Ephemeroptera (Mayflies)*: 5 families (including *Heptageniidae* flathead and *Baetidae* minnow mayflies).
- *Plecoptera (Stoneflies)*: 3 families (including *Perlidae* common stoneflies, proving year-round cold, highly oxygenated water).
- *Trichoptera (Caddisflies)*: 6 families (including *Hydropsychidae* net-spinners and *Glossosomatidae* saddlecase makers).
- **EPT Index Evaluation**: The rapid colonization of highly sensitive EPT families proves that water chemistry, substrate cleanliness, and macro-habitat complexity have been successfully restored. This abundant insect community provides a rich, sustainable food source for the brown trout fishery.

V. Wild Brown Trout Spawning Redd Count

During our late autumn spawning survey (November 15, 2026), field biologists and UGA Warnell interns mapped the restored reach:

- **Active Spawning Redds Mapped**: **12 active brown trout redds** (nests) were physically observed and georeferenced inside our newly restored sorted gravel riffle runs!

- **Spawning Metrics**:
- *Average Redd Area*: 0.65 m² (7.0 sq ft).
- *Redd Substrate Sizing*: D■■ median gravel grain size of 22.5 mm (perfect spawning sorting).
- *Water Velocity over Redds*: Measured at an optimal 0.35 m/s to ensure continuous hyporheic downwelling flow.
- *Redd Spawning Density*: Approximately 1 active nest per 125 linear feet of stream reach, proving that the Anderson Creek geomorphic restoration serves as an active, high-functioning wild trout spawning sanctuary.

Prepared by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin.